Cambridge (CIE) A Level Maths: Pure 1

Your notes

Basic Trigonometry

Contents

- Definitions of Trigonometric Functions
- Right-Angled Triangles
- Non-Right-Angled Triangles

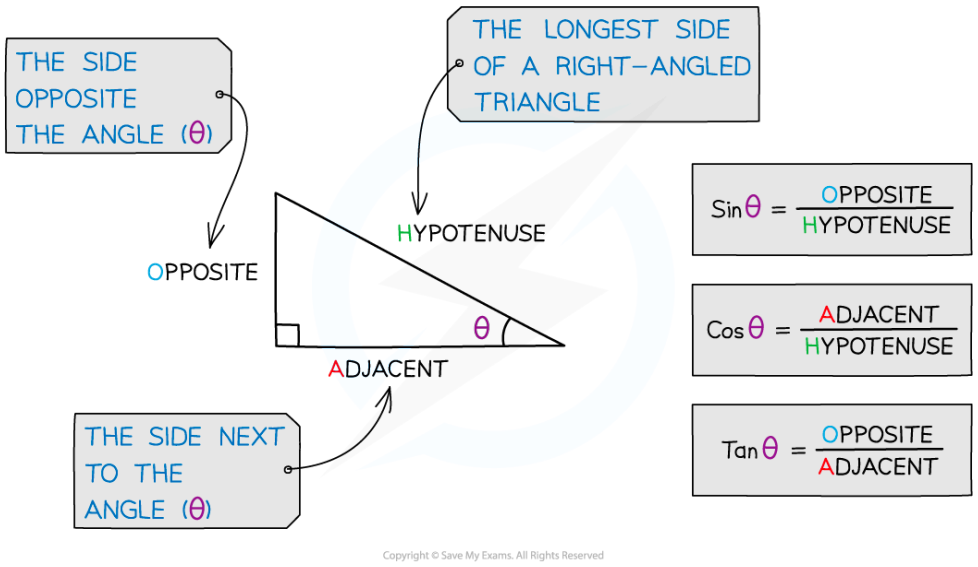
© 2025 Save My Exams, Ltd.

Your notes

Definitions of trigonometric functions

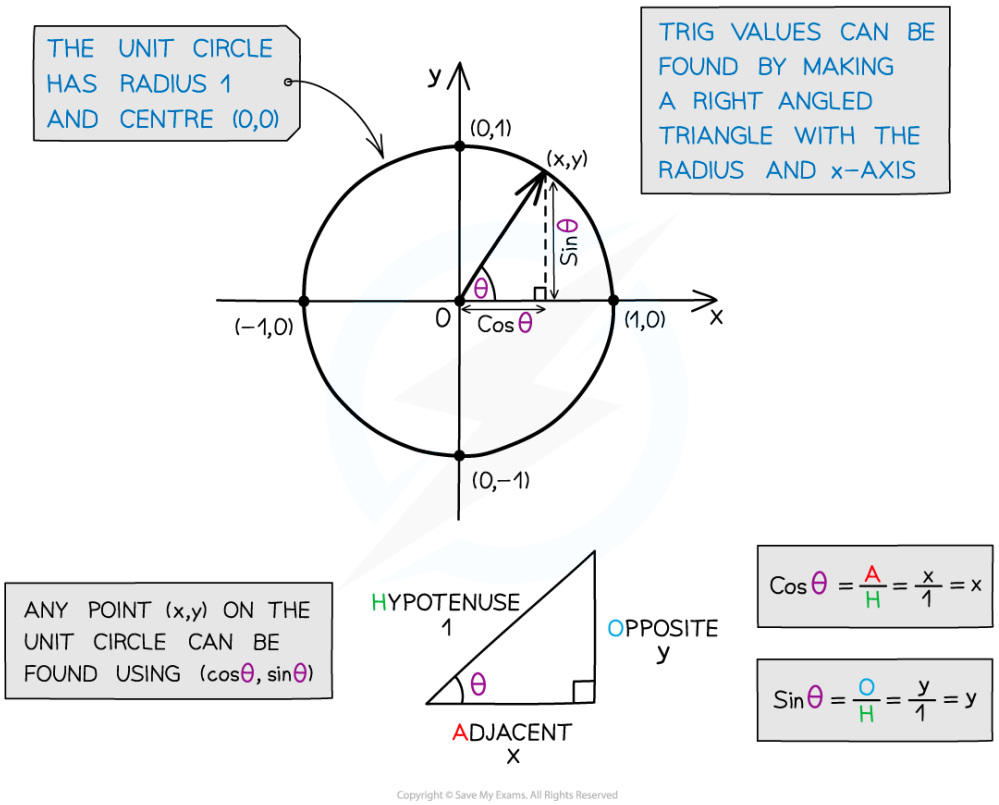
What is trigonometry?

- Trigonometry looks at the relationship between side lengths and angles of triangles
- It comes from the Greek words trigonon meaning 'triangle' and metron meaning 'measure'
- The three trigonometric functions Sine, Cosine and Tangent come from ratios of side lengths in right angled triangles



What is the unit circle in trigonometry?

- The unit circle is a circle with radius 1 and centre (0, 0)
- It can be used to calculate trig values as a co-ordinate point (x, y) on the circle
- (x, y) = (cos θ, sin θ), where θ is the angle measured anticlockwise from the positive x-axis
- It allows us to calculate **sin**, **cos** and **tan** for angles greater than 90°



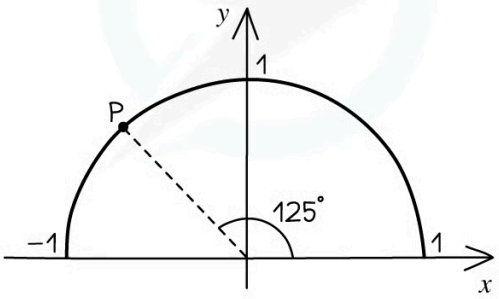
Your notes

Worked Example

Your notes

a) The co-ordinates of a point on the unit circle, to 3 s.f., are (0.629, 0.777). Find θ to the nearest degree.

b) Find the coordinates of point P on the unit circle. Give your answer to 2 d.p.



Copyright © Save My Exams. All Rights Reserved
save my exams

WE KNOW $(x,y) = (\cos\theta, \sin\theta)$
SO $\cos\theta = 0.629$ AND $\sin\theta = 0.777$

a) $\cos\theta = 0.629$

$\theta = \cos^{-1}(0.629) = 51°$ (TO THE NEAREST DEGREE)

OR

$\sin\theta = 0.777$

$\theta = \sin^{-1}(0.777) = 51°$ (TO THE NEAREST DEGREE)

BOTH WORK BUT YOU ONLY NEED TO USE ONE

b) $\cos(125) = -0.5735764...$

$\sin(125) = 0.8191520...$

SO $P = (-0.57, 0.82)$

AS P IS ON THE UNIT CIRCLE WE KNOW $(x,y) = (\cos\theta, \sin\theta)$

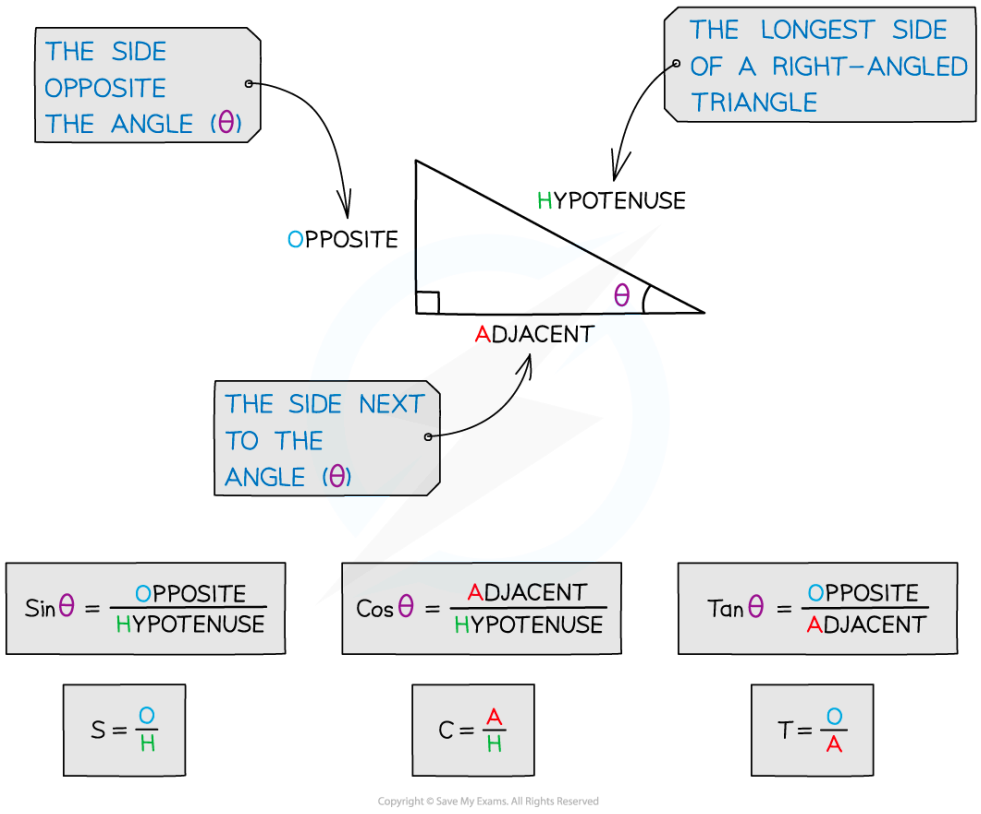
Copyright © Save My Exams. All Rights Reserved
save my exams

Your notes

Right-angled triangles

What is SOHCAHTOA?

- SOH, CAH, and TOA are an easy way to remember the three basic trigonometry ratios
- They show the relationship between an angle and the different sides of a right-angled triangle which we can label **Hypotenuse**, **Adjacent** and **Opposite**
- We use them to find missing angles and sides for right-angled triangles
  - **Hypotenuse** is the longest side of a right-angled triangle
  - **Adjacent** is the side next to the angle you are working with (in between the angle and the right angle)
  - **Opposite** is the side opposite the angle you are working with



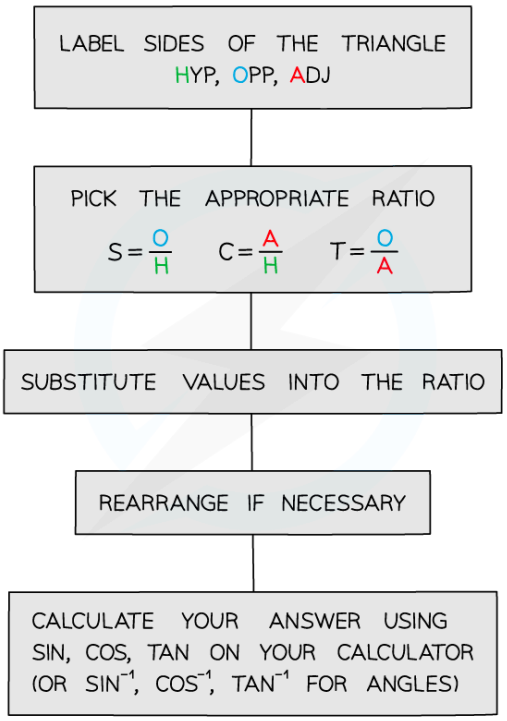
- We use the inverse trig functions to find missing angles

$\theta = \sin^{-1}\left(\frac{O}{H}\right)$ $\theta = \cos^{-1}\left(\frac{A}{H}\right)$ $\theta = \tan^{-1}\left(\frac{O}{A}\right)$

Copyright © Save My Exams. All Rights Reserved

Your notes

How do I use SOHCAHTOA?



Copyright © Save My Exams. All Rights Reserved

Examiner Tips and Tricks

- Remember SOHCAHTOA are only for calculating missing sides and angles for right-angled triangles
  - For non-right-angled triangles you will need to use the Sine or Cosine rules.
- Make sure your calculator is in degree mode (D).
- Always check your answers are sensible…
  - Should my answer be longer or shorter than the sides I have already i.e. is it the hypotenuse?
  - Is my angle sensible…
  - Have I remembered to use $^{-1}$?

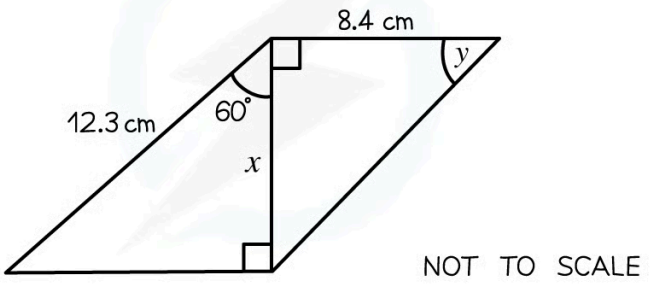
- Check if the question asks you to round your answer.

Your notes

Worked Example

Find the values of x and y in the following diagram.

Give your answers to 3 significant figures.

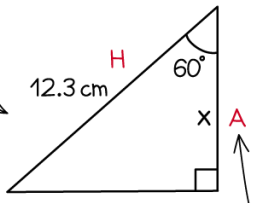


Copyright © Save My Exams. All Rights Reserved
save my exams

Your notes

TO FIND x

SKETCH AND LABEL THE TRIANGLE



WE KNOW **H** AND WE WANT TO FIND **A** SO MUST NEED **CAH**

$\cos\theta = \frac{A}{H}$

$\cos 60 = \frac{x}{12.3}$

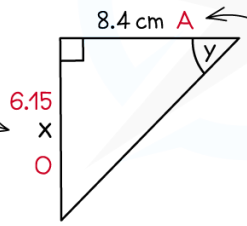
SUBSTITUTE VALUES AND REARRANGE

$x = 12.3 \times \cos 60$

$x = 6.15\text{ cm}$

TO FIND y

NOW WE'VE FOUND x=6.15, SKETCH AND LABEL THE TRIANGLE



WE HAVE **A** AND **O** SO MUST NEED **TOA**

$\tan\theta = \frac{O}{A}$

$\tan y = \frac{6.15}{8.4}$

USE $\tan^{-1}$ TO CALCULATE THE ANGLE

$y = \tan^{-1}\left(\frac{6.15}{8.4}\right)$

$y = 36.209...$

$y = 36.2$

Copyright © Save My Exams. All Rights Reserved
save my exams

Your notes

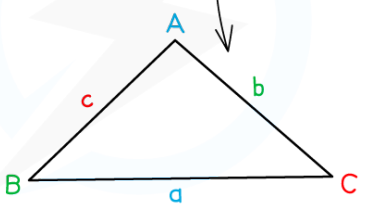
Non-right-angled triangles

What is the sine rule, cosine rule and area rule for non-right-angled triangles?

- The Sine Rule, Cosine Rule and Area Formula can be used for ANY type of triangle
- They can help us calculate missing side lengths, angles and area

ALWAYS RELABEL YOUR TRIANGLE ABC/abc TO MATCH THE RULE YOU NEED

LABEL YOUR TRIANGLE WITH CAPITALS FOR ANGLES AND LOWER CASE FOR THE OPPOSITE SIDE



Copyright © Save My Exams. All Rights Reserved

TO FIND AN UNKNOWN SIDE PUT a, b, c ON TOP

SINE RULE

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

OR

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

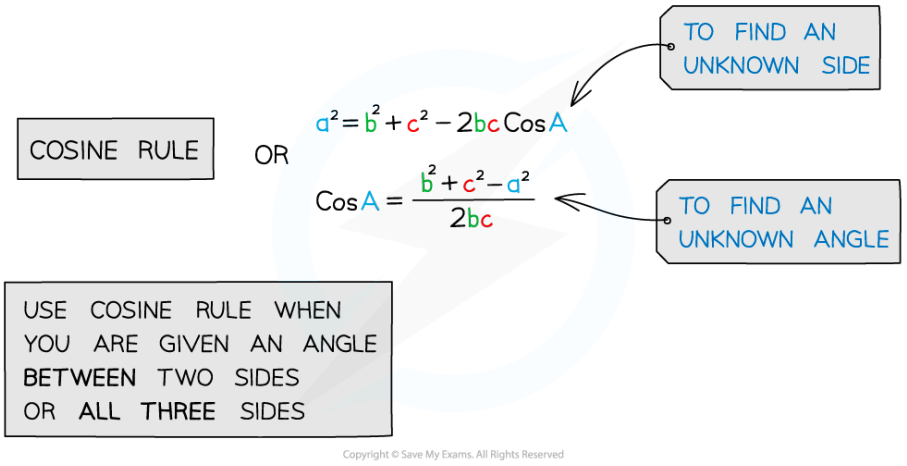
USE SINE RULE WHEN YOU HAVE ANY **OPPOSITE PAIRS** OF SIDES AND ANGLES

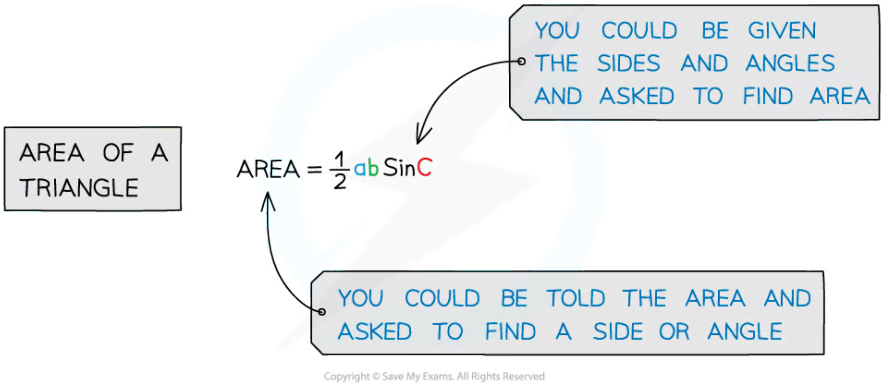
TO FIND AN UNKNOWN ANGLE PUT Sin A, Sin B, Sin C ON TOP

BE CAREFUL! SINE RULE WILL **ALWAYS** FIND THE ACUTE ANGLE. TO FIND THE OBTUSE ANGLE SUBTRACT FROM 180°

Copyright © Save My Exams. All Rights Reserved

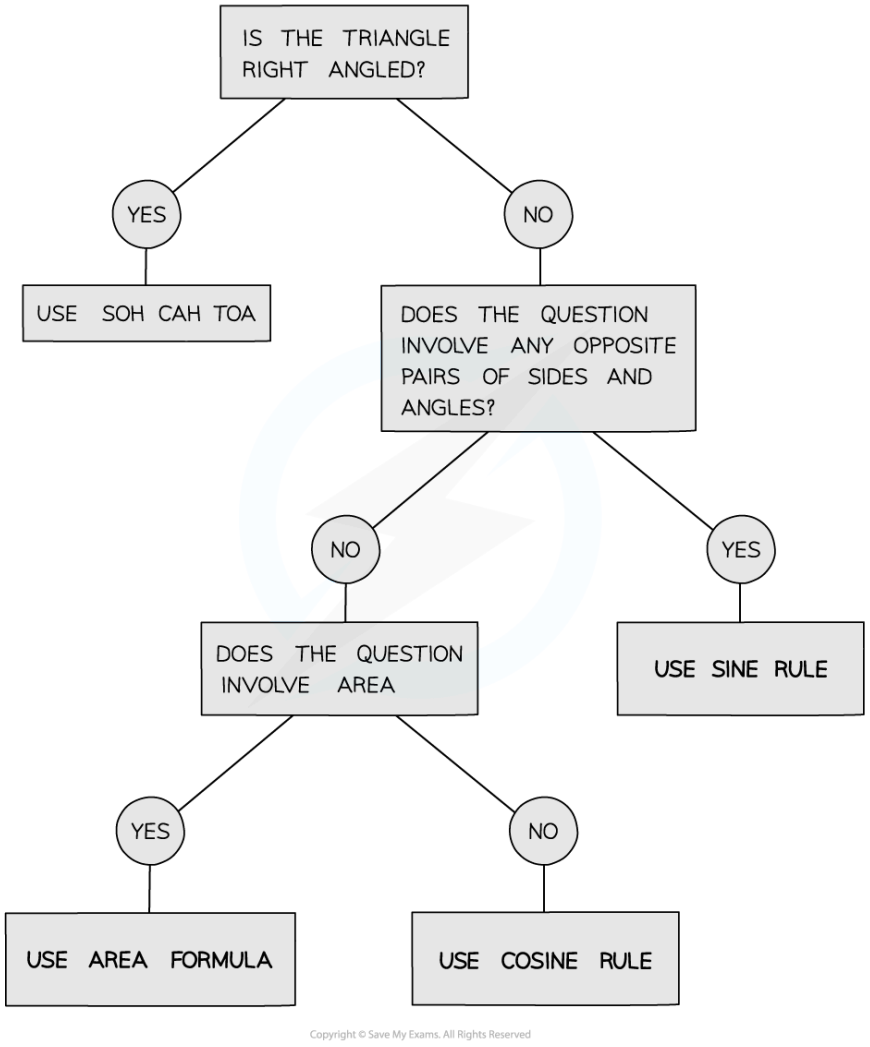
Your notes





How do I know which rule to use?

Your notes



- Once you know which rule to use, simply label your triangle, substitute in the values and rearrange if necessary

Examiner Tips and Tricks

- If you are given two angles in a triangle, finding the missing angle by subtracting from 180° might help.
- Make sure you are careful to pick the correct rule to use, you'll need to remember all the rules as they aren't in the formula booklet.
- Make sure your calculator is in degree mode (D).
- Always check your answers are sensible…
  - Have I remembered to square root at the end?
  - Have I remembered to use $^{-1}$?
- Check if the question asks you to round your answer.

Your notes

Worked Example

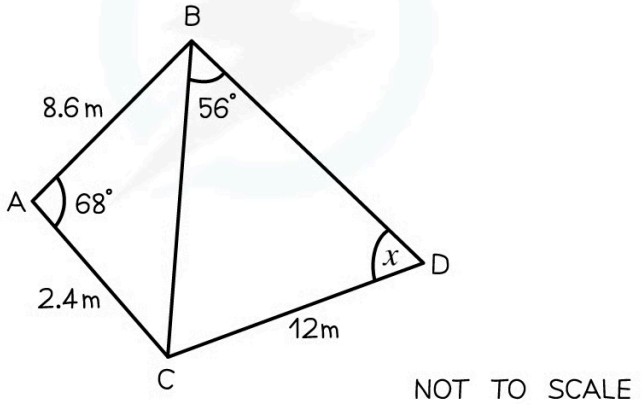
ABCD is a quadrilateral as shown in the diagram below.

a) Find the area of ABC.

Angle x is an acute angle.

b) Work out the size of angle x.

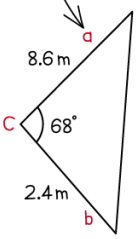
Give your answers to 3 significant figures.



Copyright © Save My Exams. All Rights Reserved
save my exams

Your notes

a)



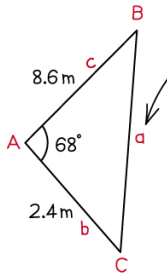
IT DOESN'T MATTER WHICH WAY AROUND YOU LABEL a AND b

RELABEL DIAGRAM TO MAKE THE GIVEN ANGLE C

SUBSTITUTE AND CALCULATE

$$\text{AREA} = \frac{1}{2}ab\sin C$$
$$= \frac{1}{2} \times 8.6 \times 2.4 \times \sin 68$$
$$= 9.568...$$
$$= 9.57\,m^2 \quad (3SF)$$

b)

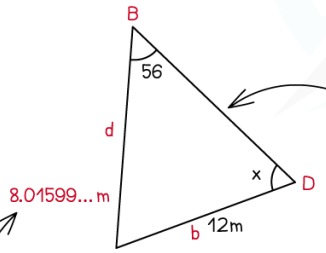


WE HAVE AN ANGLE BETWEEN TWO SIDES SO WE CAN START BY FINDING THE LENGTH OF BC USING COSINE RULE

$$a^2 = b^2 + c^2 - 2bc\cos A$$
$$a^2 = 2.4^2 + 8.6^2 - 2 \times 2.4 \times 8.6\cos 68$$
$$a^2 = 64.256...$$

DON'T FORGET TO $\sqrt{\text{SQUARE ROOT}}$

$$a = \sqrt{64.256...} = 8.01599...$$



WE NOW HAVE OPPOSITE PAIRS OF SIDES AND ANGLES SO CAN USE SINE RULE TO FIND x

THIS IS WHAT WE HAVE JUST FOUND

$$\frac{\sin D}{d} = \frac{\sin B}{b}$$

$$\frac{\sin x}{8.01599} = \frac{\sin 56}{12}$$

USE $\boxed{\text{ANS}}$ BUTTON ON YOUR CALCULATOR TO PRESERVE ACCURACY

$$\sin x = 8.01599... \times \frac{\sin 56}{12}$$

$$\sin x = 0.55379...$$

USE INVERSE^{-1} TO FIND x

$$x = \sin^{-1}(0.55379...)$$

$$x = 33.627...$$

$x = 33.6^\circ$ (3SF)

Copyright © Save My Exams. All Rights Reserved

Your notes